MIL-G-50582 (MU)
3 September 1970
SUPERSEDING
(See 6.5)

MILITARY SPECIFICATION

GRENADES, LOWER SHELL

AND

SHELL-TUBE ASSEMBLY FOR

1. SCOPE

1.1 This specification covers the metal parts for Shell-Tube Assembly and Lower Shell (Dwgs. D-8865569 and D-8865570) for use with various grenades.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-A-2550	-	Ammunition and Special Weapons, General Specification for
MIL-B-117	-	Bags and Sleeves, Interior Packaging
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)

FSC: 1330

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MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-1168	-	Lot Numbering of Ammunition
MIL-STD-1169	-	Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235	-	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

US ARMY MUNITIONS COMMAND

D-8865569	-	Shell, Lower
D-8865570	-	Shell-Tube Assembly

PUBLICATIONS

ARMY

ET-8865570	-	Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM B 117-64	-	Standard Method of Salt Spray (Fog) Testing

(Application for copies should be addressed to: American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103).

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3. REQUIREMENTS

3.1 Materials.-Materials shall be in accordance with the applicable drawings and specifications.

3.2 Parts and assemblies.-The lower shell and the shell-tube assembly shall comply with all requirements specified on Drawings (Dwgs) D-8865569 and D-8865570, respectively, all associated drawings, and with all requirements specified in applicable specifications.

3.3 Workmanship.-All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products, and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents.

3.4 Submission of results of contractor-conducted examinations and tests.-The contractor shall forward copies of the results of any of his examinations or tests to the designated address as required by the contract. (See 6.4)

3.5 First article testing.-This specification makes provisions for first article testing. Submission of first article quantity by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

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4.1.1 Submission of product.-At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection.

4.2.1 Submission.-The contractor shall submit a first article quantity as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.2.2. (See 6.1) The first article sample shall consist of the following:

a. Seventy-five (75) completed shell - tube assemblies.
b. Seventy-five (75) subassemblies.
c. Seventy-five (75) of each part.
d. Two (2) stock specimens, each comprised of the width of the stock to be used for production, and the length, equivalent to the circumference of the tool to be utilized for production, shall be selected from the coil of stock prior to forming the upper shell and lower shell. One sample shall be taken from the beginning of the coil; the other sample shall be taken from the end of the coil. Each specimen shall be identified for the direction of rolling and identified with an index position of the tool. The first article sample shall be submitted and approved prior to initiation of regular production.

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All parts and completed assemblies submitted shall have been produced by the contractor (or furnished by a supplier) using the same production processes, procedures and equipment as will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of conformance. A first article quantity, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification, or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 Inspections to be performed.-Assemblies and components will be subjected by the Government to any or all the examinations or tests specified in 4.3.2 and 4.3.3 of this specification, and any or all requirements of the applicable drawings. In addition, any or all of the following examinations and tests will be performed.

4.2.2.1 Quantities for first article tests.-After classification of defects inspection and other inspections have been completed as specified in 4.2.2, the following quantities shall be subjected to the following tests:

4.2.2.1.1 Salt spray test of upper shell and lower shell (See Dwgs. D-8865568 and D-8865569, respectively). Twenty-five (25) each upper shells and lower shells shall be subjected to this test. The test shall be performed as specified in 4.4.2.

4.2.2.1.2 Push-through test of spring grip fastener of shell-tube assembly (See Dwg. D-8865570).-The spring grip fasteners of the seventy-five (75) shell-tube assemblies shall be subjected to this test. The test shall be performed as specified in 4.4.3.

4.2.2.1.3 Minimum inside volume of hemispheres (prior to phosphating) (See Dwgs D-8865568 and D-8865569).-Ten (10) each of upper shells and lower shells shall be subjected to this test. This determination shall be performed as specified in 4.4.4.

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4.2.2.1.4 Dimensional inspection of stock specimens (See Dwgs D-8865568 and D-8865569).-The two (2) stock specimens (See 4.2.1 d) shall be subjected to the dimensional inspection specified in 4.4.1.

4.2.3 Rejection.-If any assembly, component or stock specimen fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly, component, or stock specimen in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article quantity or portion thereof. Until a first article quantity is accepted, the contractor is in no way authorized by the Government to resume regular production unless otherwise directed by the Contracting Officer.

4.3 Inspection provisions.

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with the requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (See 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling

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plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. In addition, at the option of the procuring activity, defect characteristics of the same classification which have been separated into two (2) groups to specifically allow the optional method of inspection in accordance with 4.3.3.5 and 4.3.3.6, may be combined into one (1) group when the option of 4.3.3.5 and 4.3.3.6 is not exercised. The AQL for the combined grouping will be as indicated in the applicable classification of defects paragraph. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Sleeve, detonator tube (See Dwg. B-8864980 covering a detail of Dwg. D-8865570).

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:	None defined		
Major	*AQL 0.40 percent (See 4.3.3.5)		
101.	Diameter of body	Gage	01001
Major	AQL 0.40 Percent		
102.	Burr	Visual	01002
103.	Crack or perforation	Visual	01003
Minor	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.3)	Visual	01004

*Combined AQL for Major defects 101-103, inclusive, is as follows: AQL 0.65 Percent (See 4.3.2)

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4.3.2.2 Tube, detonator (See Dwg. C-8865558 covering a detail of Dwg. D-8865570).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 1.00 Percent		
101.	Location of flash hole	Gage	02001
102.	Diameter of flash hole	Gage	02002
103.	Pitch diameter of thread	Gage	02003
104.	Diameter of cavity, above thread, min.	Gage	02004
105.	Concentricity of diameter of cavity above thread with outside diameter	Gage	02005
106.	Concentricity of smaller (sleeve) cavity with outside diameter	Gage	02006
107.	Length of effective thread, min.	Gage	02007
108.	Angle missing from flash hole	Visual	02008
109.	Flash hole missing	Visual	02009
Minor	AQL 1.50 Percent		
201.	Minor diameter of thread, max.	Gage	02010
202.	Diameter of thread counterbore	Gage	02011
203.	Depth of thread counterbore	Gage	02012
204.	Evidence of poor workmanship (See 3.3)	Visual	02013

4.3.2.3 Shell, lower (See Dwg. D-8865569).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 1.00 Percent		
101.	Largest outside diameter	Gage	03001
102.	Thickness, Min.	Gage	03002
103.	Contour improper	Gage	03003
104.	Diameter of hole	Gage	03004
105.	Concentricity of hole with largest outside diameter including perpendicularity	Gage	03005
106.	Angle incorrect on lip end	Visual	03006

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Categories	Defects	Method of Inspection	Code No.
Minor	AQL 1.00 Percent		
201.	Evidence of poor workmanship (See 3.3)	Visual	03007
202.	Small lip radius missing	Visual	03008

4.3.2.4 Shell, upper (See Dwg. D-8865568).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 1.00 Percent		
101.	Distance from hole face to flat end including chamfer	Gage	04001
102.	Outside diameter	Gage	04002
103.	Diameter of hole, min.	Gage	04003
104.	Depth to hole face, min.	Gage	04004
105.	Concentricity of hole with outside diameter	Gage	04005
106.	Concentricity of second largest outside diameter with largest outside diameter	Gage	04006
107.	Thickness, min.	Gage	04007
108.	Contour improper	Gage	04008
Major	*AQL 0.65 Percent (See 4.3.3.5)		
109.	Angle of outside diameter incorrect	Gage	04009
Minor	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.3)	Visual	04010

*Combined AQL for Major defects 101-109, inclusive, is as follows: AQL 1.00 Percent (See 4.3.2)

4.3.2.5 Shell-tube assembly (See Dwg. D-8865570).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	*AQL 0.40 Percent (See 4.3.3.6)		
101.	Torque test	Test	05001

*Combined AQL for Major defects 101-108, inclusive, is as follows: AQL 1.00 Percent (See 4.3.2)

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Major	AQL 1.00 Percent		
102.	Protrusion of tube assembly	Gage	05002
103.	Concentricity of diameter of tube assembly with outside diameter of upper shell including perpendicularity with shell face	Gage	05003
104.	Inside diameter of spring grip fastener, max.	Gage	05004
105.	Outside diameter of tube assembly at threaded end	Gage	05005
106.	Inside diameter of detonator tube sleeve, min.	Gage	05006
107.	Distance between shell hemisphere face and tube face	Gage	05007
108.	Component or operation missing	Visual	05008
Minor	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.3)	Visual	05009

4.3.3 Testing.

4.3.3.1 Dimensional inspection of stock specimen (See Dwgs. D-8865568 and D-8865569) Major defects, Code No. 06001. Two (2) stock specimens shall be taken from each coil of finished stock just prior to forming the upper and lower shells. The width of each specimen shall be the width of the stock, and the length shall be as necessary to obtain the total surface area of the tool to be used in production. One (1) specimen shall be taken from the beginning of the coil and the other specimen shall be taken from the end of the coil. An additional specimen shall be taken whenever a new, repaired, or altered tool is entered into production. Each specimen shall be identified with an index position on the tool and for the direction of travel of the stock with respect to the tool. The specimens shall also be identified as being from the beginning or end of the coil. If, for any specimen, the average value of fifteen (15) measurements for the total thickness dimension and the depth dimension fails to comply, or if the direction of tool travel does not comply with the applicable drawing requirements, the coil, or portion thereof, from which the specimen was taken, shall be rejected. In addition, the tool which produced the rejected specimen shall be removed from production. All shells produced from the rejected coil shall be rejected. The dimensional inspection shall be performed as specified in 4.4.1 using equipment in accordance with 4.3.4.

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The specimen from a new, repaired, or adjusted tool entering production shall be subjected to dimensional inspection of all stock dimensions specified on the applicable drawings. Fifteen (15) readings shall be taken for each dimension. The tool shall not be used in production until dimensional compliance with the drawing requirements has been determined from the specimen produced by the tool.

4.3.3.2 Salt spray test of upper shell and lower shell (See Dwgs. D-8865568 and D-8865569, respectively) Major defect, Code No. 07001).

4.3.3.2.1 Phosphate - coated shells (batch-type process). Fifteen (15) each of upper shell and lower shell shall be selected from each batch and subjected to the salt spray test as specified on the applicable drawing. If any part fails to comply with the specified requirement, the batch represented by the failing part shall be rejected. The test shall be performed as specified in 4.4.2.

4.3.3.2.2 Phosphate - coated shells (continuous-type process). Fifteen (15) each of upper shell and lower shell shall be selected from each four (4) hours of production, or portion thereof, and subjected to the salt spray test as specified on the applicable drawing. If any part fails to comply with the specification requirement, the quantity represented by the failing part shall be rejected. The test shall be performed as specified in 4.4.2.

4.3.3.2.3 Phosphate - coated shells (lot plan).-Twenty-five (25) each of upper shell and lower shell shall be selected from each lot and subjected to the salt spray test as specified on the applicable drawing. If any part fails to comply with the specified requirement, the lot shall be rejected. The test shall be performed as specified in 4.4.2.

4.3.3.2.4 Plated shells (when applicable).-Twenty-five (25) each of upper shell and lower shell shall be selected from each lot and subjected to the salt spray test as specified on the applicable drawing. If any part fails to comply with the specified requirement, the lot shall be rejected. The test shall be performed as specified in 4.4.2.

4.3.3.3 Push-through test of spring grip fastener of shell-tube assembly (See Dwg. D-8865570) Major defect, Code No. 08001. One hundred twenty-five (125) shell- tube assemblies shall be selected from each lot and subjected to this test. If two (2) or more assemblies fail to comply with the requirement specified on the applicable drawing, the lot shall be rejected. The test shall be performed as specified in 4.4.3 using equipment in accordance with 4.3.4.

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4.3.3.4 Minimum inside volume of hemispheres (prior to phosphating) (See Dwgs. D-8865568 and D-8865569) Major defect, Code No. 09001.-A sample of ten (10) each upper shells and lower shells shall be selected for determination of minimum volume requirement to qualify the production process. In addition, five (5) parts shall be selected from the last portion of each four (4) hours production or fraction thereof, for determination of minimum volume requirement as a control of the production process. This determination shall be performed as specified in 4.4.4 using equipment in accordance with 4.3.4. If any defective parts are found during qualification of the production process, the process producing the defective part shall not be used in production. If any defective parts are found when inspection is performed for control of the production process, the process producing the defective part shall be removed from production. Further, that portion of production since the last production process control check shall be returned to the contractor for corrective action.

All production processes removed from production because of some fault may, after corrective action, be returned to production, providing they pass the qualification test above. The contractor may request a change of inspection frequency, providing he presents objective evidence to the Contracting Officer to substantiate the request.

4.3.3.5 Dimensional control of stamped parts.-In place of the normal sampling associated with the classification of defects, and with the approval of the Contracting Officer, a sample of at least ten (10) parts may be dimensionally inspected to qualify the tool used in the stamping process for use in production. In addition, a random sample of five (5) parts shall be selected from the last portion of each hour's production for dimensional inspection as a control of the tool during production.

If any defective parts are found during qualification of the tool, the tool producing the defective part shall not be used in production.

If any defective parts are found when inspection is performed for control of the tool, the tool producing the defective part shall be removed from production. Further, that portion of production since the last tool control check shall be returned to the contractor for corrective action.

All tools removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the Contracting Officer to substantiate the request.

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4.3.3.6 Minimum torque of tube assembly (See Dwg. D-8865570), Major defect, Code No. 05001.-In place of the normal sampling procedure associated with the classification of defects, and with the approval of the Contracting Officer, the following inspection procedures may be used: at the start of each day, and whenever a new piece of equipment is put into operation, five (5) tube assemblies shall be selected from each piece of equipment for determination of minimum assembly torque of the tube assembly for qualifying the equipment. In addition, five (5) assemblies per machine shall be selected from each hour's production, or fraction thereof, for the determination of the minimum tube assembly torque as a control of the equipment. If any defective assemblies are found during qualification of the equipment, the equipment producing the defective assemblies shall not be used in production. If any defective assembly is found when inspection is performed for control of the equipment, the equipment having produced the defective assembly shall be removed from production.

Further, that portion of production since the last equipment check shall be returned to the contractor for corrective action. All equipment removed from production because of some fault may, after corrective action, be returned to production providing it passes the qualification test above.

4.3.4 Inspection equipment.-Equipment Tabulation (ET) number 8865570 and Equipment Lists (EL's) referenced thereon identify the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3. Supply and maintenance of the equipment listed on the EL's shall be in accordance with MIL-I-45607.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of this specification may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey, and whenever inspection equipment drawings are needed.

4.4 Test methods and procedures.

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4.4.1 Dimensional inspection of stock specimens. Fifteen (15) measurements shall be taken on each sample specimen for the total thickness dimension and the depth dimension. There shall be three (3) points of measurement across the width of the specimen at five (5) places along the length. The spacing of the readings shall be approximately uniform. One measurement for each characteristic shall be performed at each of the fifteen (15) measuring points on the specimen. An average value for each characteristic shall be computed for each specimen for which these measurements were obtained. In addition to the above, stock specimens from a new, repaired, or adjusted tool entering production shall have the following characteristics measured: groove radius, spherical radius, and the .0965 dimension.

4.4.2 Salt spray test of upper shell and lower shell. The upper and lower shells shall be subjected to the salt spray test in accordance with ASTM Standard Method B 117-64. The sodium chloride solution shall be as indicated on the applicable drawing. This is a destructive test. Parts so tested shall be discarded.

4.4.3 Push-through test of spring grip fastener of shell-tube assembly.-The shell-tube assembly shall be subjected to this test using the fixture and procedure specified on the Equipment List. Observation shall be made for compliance with the requirement specified on the applicable drawing. This is a destructive test. Assemblies so tested shall not be returned to the lot.

4.4.4 Minimum inside volume of hemispheres (prior to phosphating).-The upper and lower shells shall be subjected to this determination using equipment and procedures specified on the Equipment List.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging (Level C).

5.1.1 Lower shell (Dwg. D-8865569).-The lower shell shall be packed in accordance with MIL-STD-1169, except that a Type 1, Class E, Spec MIL-B-117 barrier bag and an outer fiber box shall be used in addition to the inner fiber box. The inner fiber box corners shall be blunted before placing in bag. Prior to sealing bag, exhaust all excess air. Maximum moisture content of inner fiber box and any other packaging material used within the inner box shall be 9%.

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5.1.2 Shell-tube assembly (Dwg. D-8865570).-Packaging shall be in accordance with 5.1.1.

5.2 Packing (Level C).

5.2.1 Lower shell (Dwg. D-8865569).-Packing shall be in accordance with 5.1.1.

5.2.2 Shell-tube assembly (Dwg. D-8865570).-Packing shall be in accordance with 5.1.1.

5.3 Marking.-Marking shall be in accordance with MIL-STD-1169.

5.4 Shipping.-When components from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number, and date of this specification.
b. Data cards shall be prepared for each lot in accordance with the information specified in MIL-STD-1167.
c. Provisions for submission of first article samples.

6.2 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. These code numbers shall be listed on drawings of equipment designs submitted by the contractor (See 6.3.5).

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Equipment lists.-Equipment lists indicate the availability of inspection equipment designs by showing in the "Part Number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated

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in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the "Part Number" column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications: Mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, with the approval of the Government, design alternate inspection equipment to facilitate his operations. Such contractor-prepared designs (See 6.3.4) or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finishes, notes and operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design

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procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to Specification MIL-D-1000, Category A, Form 3. Legibility and reproducibility shall permit conventional making of clearly understandable, high-contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor design.-All submitted designs shall contain a reference to the applicable 5-digit code number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND3. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract. Submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished, normally, within one month after receipt.

6.4 Submission of results of contractor-conducted examinations and tests.-Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND3.

6.5 Supersession data.-This specification includes the requirements of, and supersedes, Purchase Specification AEI-160E, dated 12 June 1968.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number: 1330-A221

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